

Mid-Way Report

Submission for Feedback

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Dataset

We plan to use the [NBA Traditional Statistics Dataset](#), which contains detailed player and team performance data across NBA seasons from 1997-2025. The dataset is extracted from the official NBA statistics database and includes comprehensive records for both individual players and teams.

The data captures a range of features, including both, player and team names, games played, minutes, points, assists, rebounds, steals, blocks, shooting percentages, and other essential key performance indicators. This level of granularity makes the dataset highly suitable for both possibilities of insightful and predictive analytics.

After a visual oversight, we can see that cleaning will be required, such as handling missing or inconsistent records, possible standardizing of column types and potential duplications between player and team files. Providing an excellent opportunity to refine the dataset into a dense and high-quality resource for subsequent analysis.

Technology

Apache Pig – For data cleaning operations such as handling NULL or missing values and ensuring correct data types across both player and team datasets.

Apache Spark – For large-scale data processing, enabling efficient computation of team and player statistics, trends, and model training using SparkML for predictive analytics.

Apache Hive – For executing SQL-like queries on cleaned datasets stored in HDFS, making aggregation and filtering operations easier.

Matplotlib – For generating visualizations that depict team and player performance, efficiency comparisons, and season-over-season trends.

Analytics

We plan to explore a combination of descriptive, comparative, and predictive analyses using the NBA dataset. Key analytical areas include:

- **Player Efficiency:** Analyse player performance by discovering their percentages across a range of metrics throughout the season.
- **Team Performance:** Identify teams with the highest and lowest percentages across defensive and offensive metrics to see the standards of the best and worst teams.
- **Temporal Trends:** Track performance metrics across different seasons to identify trends from particular players and teams.
- **Predictive Modelling:** Create a model to predict team success based on key performance indicators, as well as player efficiency.

- **Hidden Performers:** Acknowledge underrated players or teams that outperform expectations based on performance metrics and statistics.

The overall goal is to better understand what statistical factors and patterns that contribute most to team success and individual player performance in the NBA. Analyzing the interplay between scoring, efficiency, and defensive stats may reveal some hidden strategic trends that distinguish championship quality teams amongst the average performers.

Team Roles and Tasks

Beblin Team Roles

Resource Investigator - <i>AD</i>	Specialist - <i>BC</i>	Completer Finisher - <i>BC</i>
Plant - <i>BC</i>	Shaper - <i>AD</i>	Coordinator - <i>LH</i>
Monitor Evaluator - <i>LH</i>	Implementer - <i>LH</i>	Teamworker - <i>AD</i>

Specific Tasks

- Inspect and clean datasets for missing / inconsistent data.
- Execute queries using Apache Hive, Pig, and Spark to derive meaningful aggregations and comparisons.
- Perform data transformation and quality assurance in Apache Pig.
- Develop clear visualizations using Matplotlib to present player and team-level insights.
- Build a simple predictive model using SparkML to forecast performance metrics.
- Compile and refine the final project report for submission.
- Maintain version control and updates via GitLab.